

Effect of a youth-led early childhood care and education programme on children's development and learning in rural Sindh, Pakistan (LEAPS): a stepped-wedge cluster-randomised implementation trial



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Summary

Background In low-income and middle-income countries, an estimated 181·9 million (74·6%) preschool-aged children do not receive adequate nurturing care in health, nutrition, protection, learning, and responsive care, thus jeopardising their healthy development across the life course. Working alongside the health sector, multisectoral actions including social protection and education are necessary to achieve child health and development outcomes. Innovations are needed to expand access to high-quality early childhood care and education (ECCE) for young children and opportunities for youth development. Youth Leaders for Early Childhood Assuring Children are Prepared for School (LEAPS) is a two-generation programme that trains female youth aged 18–24 years to deliver ECCE. We evaluated the effectiveness of LEAPS to improve children's school readiness when delivered at scale in rural Pakistan.

Methods We implemented a stepped-wedge cluster-randomised trial with three steps from Dec 3, 2018, to June 30, 2021. 99 villages (clusters) in four districts in rural Sindh, Pakistan, were randomly assigned (1:1:1) to introduce LEAPS across three steps. Eligible clusters were those that had a feeder primary school run by the National Commission for Human Development, a department of the Ministry of Federal Education and Professional Training; were safe; had space for a LEAPS preschool; could identify a female youth to deliver the ECCE service; and had not previously participated in the pilot study. Government partners trained female youth, aged 18–24 years, to provide community-based ECCE, enrolling up to 20 children, aged 3·5–5·0 years, per class. Population-based cross-sectional surveys were conducted at baseline and after each step for children who were eligible if they resided in the cluster, were aged 4·5–5·5 years at the time of the survey, and without any severe clinical health conditions or disability. The primary outcome was children's school readiness using the International Development and Early Learning Assessment (IDELA) composite score comprising emergent numeracy, emergent literacy, socio-emotional development, and motor skills. An intention-to-treat analysis was conducted, using linear mixed models accounting for clustering and the stepped-wedge design. The trial is registered with ClinicalTrials.gov (NCT03764436).

Findings LEAPS was implemented in 91 of 99 villages. In eight clusters, a LEAPS preschool could not be set up. For the intervention programme, the average enrolment of children in a LEAPS preschool was 19 (SD 3, range 12–20). A total of 3858 children (n=3852 with complete IDELA data) were assessed across the four survey rounds conducted between Jan 1, 2019, and March 31, 2021. LEAPS increased school readiness (standardised mean difference: 0·30 [95% CI 0·20–0·40]; $p<0·0001$) as compared with the control condition.

Interpretation LEAPS offers a scalable solution to expand access to high-quality ECCE and promote children's school readiness while providing employment for female youth in rural Pakistan and settings with similarly low ECCE participation.

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Introduction

An estimated 181·9 million (74·6%) preschool-aged children living in low-income and middle-income countries (LMICs) are not meeting their developmental potential due to inadequate nurturing care.¹ The second 1000 days, from ages 2–5 years, is an opportunity to intervene to promote development.¹ The Nurturing Care

Framework (NCF) provides an evidence-based roadmap to promote healthy development through multisectoral actions across the life course from conception to newborn care to infancy, and from toddlerhood to the preschool age group, middle childhood, adolescence, and adulthood.² For preschoolers, as in the first 1000 days of life, health, nutrition, stimulation, and protection

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For the Urdu translation of the abstract see Online for appendix 1

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Research in context

Evidence before this study

A multisectoral life course approach is needed to support healthy development. Specifically for the preschool age group, health, nutrition, and protection remain important, and early childhood care and education (ECCE) programmes offer additional opportunities to enhance early learning. The life course approach to health, development, and learning is also captured in the Sustainable Development Goals for good health and wellbeing (Goal 3), quality education (Goal 4), and decent work and economic growth (Goal 8) calling for particular attention to young children and youth. Therefore, improving access to care and early learning opportunities is of paramount importance for preschoolers in Pakistan and globally. We reviewed systematic reviews, meta-analyses, and individual studies on (1) ECCE and (2) youth development programmes in low-income and middle-income countries (LMICs), published between Jan 1, 2010, and March 31, 2018 (updated in April 30, 2024). Inclusion criteria for ECCE studies were interventions for children aged 3–6 years in LMICs. The inclusion criteria for youth development studies included interventions for adolescents, youth, and young adults in LMICs.

One review found that in eight of the nine included studies, children attending ECCE services had higher scores on development and learning outcomes compared with children not attending these services. A second review reported positive impacts of 31 formal and non-formal community-based preschool programmes on children's cognitive and psychosocial development. Factors such as positive teacher-child interactions, an organised classroom environment, and several age-appropriate play materials were features of effective programmes. A recent meta-analysis of child-focused educational programmes found modest-to-high effects in LMICs and estimated a minimal package of 1 year of ECCE for all children would cost on average less than 0.5% of countries' gross domestic product.

The Lancet Commission on adolescent health and wellbeing noted that the development of capabilities for positive health and wellbeing in adulthood emerge in adolescence and young

adulthood and depend upon access to opportunities and resources and reported that community-based youth development programmes resulted in positive attitudes, problem-solving abilities, enhanced self-confidence, and empowerment for youth. A systematic review of youth development programmes identified 100 programmes in 60 LMICs, but few were evaluated for effectiveness. A 2018 landscape review of youth-led ECCE programmes identified only two programmes internationally, neither of which were evaluated.

Added value of this study

To our knowledge, the Youth Leaders for Early Childhood Assuring Children are Prepared for School (LEAPS) trial is the first study that applies a randomised evaluation design to explore the effectiveness of larger scale implementation of an intergenerational programme on multiple outcomes of child development in an LMIC. The current study, a stepped-wedge cluster-randomised implementation trial, investigated the effectiveness of the LEAPS programme on young children's school readiness and executive functioning skills. Results show a small to moderate effect on children's performance on indicators of school readiness. The evaluation found the LEAPS programme benefitted children despite school closures during the trial due to the COVID-19 pandemic.

Implications of all the available evidence

The youth-led ECCE model is relevant in contexts such as rural Pakistan with a high percentage of preschool-aged children out of school and a dearth of professional development and paid employment opportunities for female youth. Our results, together with other findings and theoretical considerations, indicate that access to high-quality educational and training opportunities for children and youth in contexts such as rural Pakistan is crucial for meeting the Sustainable Development Goals. Future work might also explore adding health and nutrition interventions to ECCE platforms, such as preschool feeding programmes, to further benefit child health and development outcomes.

remain important. During this period, there is a growth in developmental skills (eg, language and social development), which continue to be refined. Therefore, young children in this developmental period continue to need stimulating and safe environments that foster their motor, cognitive, language, and social skills. Early experimental studies in child health and development have shown the importance of continuity of multiple nurturing inputs; for example, in Colombia, young children exposed to longer durations of nutrition and centre-based early learning opportunities had greater cognitive benefits.³ Another quasi-experimental study from Viet Nam reported child cognitive gains when preschoolers were exposed to an enriched centre-based early childhood care and education (ECCE) programme

following a nutrition intervention in the first 1000 days of life.⁴ However, global population data suggest that development in this period is jeopardised by inadequate inputs of nurturing care: only 33% of young children in LMICs have access to learning materials at home, 30% receive adequate home stimulation from non-maternal caregivers, 29% have access to an ECCE programme, while 86% have a healthy weight and 67% are not stunted.⁵

Although parenting programmes delivered through health, education, or social protection platforms are an important strategy to promote early child development,⁶ the few studies in LMICs that have longitudinally followed up children observe a fade out effect.⁷ In Pakistan, a 2-year follow up of a parenting trial implemented though

a community health worker programme found reduced effects on children's development and reported only a quarter of children were subsequently enrolled in an ECCE programme.⁸ Therefore, as advocated in the NCF, education and social services are crucial to supporting child development and health outcomes over the life course, particularly in contexts of multiple adversities and scarce opportunities.^{2,9} Early childhood care and education platforms also promote health and development outcomes in the life course. For example, follow up of the Carolina Abecedarian Programme in the USA found adults who had participated in an ECCE intervention had lower risks for cardiovascular and metabolic disease¹⁰ and in Türkiye, preschoolers exposed to either enriched home stimulation or ECCE, or both, benefited in cognitive development, educational achievement, employment, and life satisfaction outcomes in adulthood.¹¹ Furthermore, ECCE services, in the short-term, might offer opportunities to support young children's health and protection.¹²

ECCE services encompass a wide range of group-based programmes for young children, including home-based care, nurseries, and pre-primary education, with a shared goal to bolster children's early child development outcomes, with particular attention to the cognitive, language, and social emotional skills needed to prepare children for life-long learning, build positive relationships, and be successful at work and healthy in adulthood. In both the early childhood and late adolescence and early adulthood developmental periods, there is a marked growth in executive functioning skills in the prefrontal cortex of the brain, which support goal-oriented behaviours (eg, self-regulation, focus, planning, and reasoning). Child executive functioning, an important marker of early school readiness, is hypothesised to develop through social interactions with others who possess strong executive functioning skills; thus, caregivers with opportunities to develop their own executive functioning capacity are in turn more capable of supporting young children with skills development.¹ Therefore, interventions targeting executive functioning skills in early childhood and youth might bolster developmental gains for two-generations.

Meta-analyses of ECCE programmes find short-term and long-term cognitive and learning impacts.¹³ However, longer-term societal and economic benefits of investing in ECCE depend not only on access, but also programme quality.¹⁴ Globally, ECCE have been effective in improving young children's development and programme quality has been identified as a predictor of effectiveness.¹⁴ One impediment to expanding access to high-quality ECCE programmes is insufficient attention to the recruitment and professional development of the early childhood workforce, often poorly valued in society.¹⁵ Given the low participation of children in LMICs receiving ECCE, innovations are urgently needed to deliver high-quality ECCE programmes at

scale.¹⁶ The Sustainable Development Goals (SDGs) recognise access to ECCE programmes as a key strategy (Target 4.2) in a life course approach to promoting human development alongside youth development programmes to promote education, training, and employment for youth (Targets 4.4 and 8.6).¹⁷ Similarly to ECCE, youth development has received little investment.¹⁸ Globally, an estimated 267 million youth (22%) are unemployed, underemployed, or have insecure jobs and no access to training, with greater disparities for youth living in rural areas and for female youth.¹⁸ Strategies, informed by developmental science, which leverage interconnections across SDG targets have the potential to benefit young children and youth.¹⁹

Youth Leaders for Early Childhood Assuring Children are Prepared for School (LEAPS) is a two-generation programme developed as a response to the development, education, and training needs of young children and youth by improving access to ECCE, diversifying the early childhood workforce, and offering career and economic opportunities to youth.^{19,20} The LEAPS programme efficacy was previously evaluated in a pilot cluster-randomised controlled trial, and benefits were found on children's development and school readiness,²⁰ as well as on youth development.¹⁶ In partnership with the the National Commission for Human Development (NCHD), a government agency, we implemented a large-scale pragmatic trial of the LEAPS model in four districts of Sindh, where 84% of preschoolers are out of ECCE.²² The aim of this study was to evaluate the effectiveness of larger-scale implementation of LEAPS to improve children's school readiness and executive functioning.

Methods

Study design and participants

A detailed description of the LEAPS study can be found in the trial protocol.²¹ Briefly, female youth as community youth leaders were trained to deliver ECCE for young children in community-based preschools in rural Sindh, Pakistan. Participants were predominately Muslim (99%). Sindhi was the most common language spoken at home (58%), followed by Seraiki (20%) and Balochi (15%). In this context, there is a high youth population and a large number of young children out of ECCE (appendix 2 p 4).

We conducted a stepped-wedge cluster-randomised trial design with three steps, which was implemented in 99 village clusters in four districts (Dadu, Khairpur, Naushahro Feroze, and Sukkur) in rural Sindh.²¹ LEAPS was implemented in partnership with NCHD, an organisation that operates under the Ministry of Federal Education and Professional Training, providing primary education for children and vocational programmes for youth in remote and rural areas of Pakistan where the Ministry of Federal Education and Professional Training has limited services.²¹ LEAPS was delivered through NCHD's primary school structure,

See Online for appendix 2

Total number of village clusters	Study period				
	Step 0 (December, 2018 to February, 2019)	Step 1 (March, 2019 to August, 2019)	Step 2 (September, 2019 to August, 2020)*	Step 3 (September, 2020 to March, 2021)†	Close out (April, 2021 to June, 2021)
33					
33					
33					

Figure 1: The stepped-wedge cluster-randomised trial design

White cells represent control periods. Green cells represent intervention periods. Blue cells represent study close out phase, which is the period in which the LEAPS preschools continued to run, and children, when developmentally ready, were transitioned or graduated to the NCHD primary feeder schools. *Schools were closed from March 3, 2020, to Sept 27, 2020, due to COVID-19 shutdown. †A partial school shutdown was imposed from Nov 26, 2020, to Jan 11, 2021, in which schools were open but were operating at reduced capacity.

called feeder primary schools. The NCHD feeder primary schools (for grades 1–5) only run in villages where the Ministry of Federal Education and Professional Training, at the district level, cannot support a primary school (eg, it cannot afford a teacher salary), supporting access to basic education in communities where there is no alternative public service. In NCHD feeder primary schools, children are accommodated through multigrade classes, including modest provision of a pre-primary class, typically with first and second grade children (or with all other children in the case where there is only one teacher; appendix 2 p 4).

A village cluster was defined as the NCHD feeder primary school catchment area, as LEAPS preschools were paired with NCHD feeder primary schools to support LEAPS students graduating from preschool to enrol in primary school. Research staff worked with NCHD to identify and screen eligible village clusters. Inclusion criteria included having an operational NCHD feeder primary school. Exclusion criteria included: (1) participation in the LEAPS pilot, (2) safety and security concerns, (3) lack of appropriate community space for the preschool, (4) lack of an eligible community youth leader candidate (a female youth aged 18–24 years, with at least a grade 10 education, residing in the same or neighbouring village who had their family's support to work), or (5) closure of an NCHD feeder primary school (n=119 village clusters identified; n=20 excluded). Cluster randomisation was used because individual randomisation within a village would not be logistically feasible, and the stepped-wedge design was used for a pragmatic evaluation of LEAPS programme implementation. All village clusters began in the control condition, defined as NCHD's usual primary education services, and were randomised to roll out the LEAPS programme at one of three timepoints. The study period was 31 months, from Dec 3, 2018, to June 30, 2021 (figure 1). In the control condition, children were either out of school, were participating in an NCHD feeder primary multigrade class, or in a private school outside of the cluster.

Population-based repeated cross-sectional surveys were used to evaluate the effect of LEAPS given the focus on population-level impacts of this intervention for communities when rolling out the intervention at scale. Children were eligible for the survey if they were aged 4·5–5·5 years at the time of the survey, resided in the NCHD feeder primary school catchment areas, and did not have severe clinical health conditions or disability. Children were selected for enrolment in the population-based survey regardless of their participation in LEAPS preschools. The cross-sectional surveys were conducted at baseline before implementation of LEAPS (round one or baseline: January–April, 2019) and after each step as LEAPS was rolled out (round two: August–November, 2019; round three: February–March, 2020, and June–August, 2020; and round four or endline: December, 2020, to March, 2021). Round 3 data collection was interrupted for 3 months (March–June, 2020) due to the COVID-19 pandemic. Assessors enrolled up to 11 child-caregiver dyads in each of the 99 village clusters. It was possible for a child to be sampled more than once in a cross-sectional survey and 923 children have been sampled more than once across the four rounds. Adult caregivers provided both written and oral informed consent, and children were asked to verbally assent. Ethical approval was obtained from the Aga Khan University Ethics Review Committee, the Pakistan National Bioethics Committee, and the Harvard T H Chan School of Public Health Institutional Review Board. There was no data monitoring committee due to the nature of the intervention and design of the study. The trial is registered at ClinicalTrials.gov (NCT03764436).

Randomisation and masking

The randomisation sequence for intervention rollout was in three steps for the 99 village clusters and computer generated using a 1:1:1 allocation ratio with block randomization (block size of 3) stratified by district. The trial included six village clusters in Dadu, nine in Sukkur, 28 in Khairpur, and 56 in Naushahro Feroze districts. The randomisation list was created by the trial statistician (CRS) who was not directly involved in field activities.

Given the nature of the intervention, masking participants and LEAPS service providers was not possible. Masking of outcome assessors was not possible because intervention activities were visible in communities but the outcome assessors were not directly told of the intervention status of the villages. To ensure data accuracy and compliance, the data management team (SS, SH, EF, QD) kept records of participants' group assignment and was thus not masked. Primary data analysis was conducted by the team's data analyst (QD), who was not masked to group assignment. Interpretation of the primary findings by CRS and all other investigators (AKY, SB, CRR, GF, LAP) was masked.

Procedures

The LEAPS programme is described in detail in the published trial protocol and in appendix 2 (pp 5–16).²¹ To summarise, technical support for the LEAPS programme was provided by the LEAPS Intervention Support Team, comprising five female team members, all from local communities with a master's level education. In January, 2019, LEAPS Intervention Support Team members led a 5-day training-of-trainers for 50 NCHD officers. Following the training-of-trainers, the NCHD officers worked with communities to identify preschool classroom spaces made available by the community and conducted recruitment drives for community youth leaders. Potential community youth leader candidates attended recruitment workshops led by the NCHD, and if selected, community youth leaders went on to attend a 12-day basic training led by the NCHD. Each training day was 8 h, and training groups had a maximum of 15 community youth leader participants.

Following training, community youth leaders received a preschool starter kit to set up the preschool space with community support and conducted enrolment drives to register children aged 3–5 years who resided within catchment area. Each preschool classroom had a maximum teacher–child ratio of 1:20, and community youth leaders were coached to enrol an equal number of girls and boys. LEAPS preschools followed a 6-day school week, with one 3-h session per day. The LEAPS preschool curriculum is aligned with the Government of Pakistan's ECCE curriculum 2017, supporting children to complete 50 key development indicators across seven learning areas (inspired from High Scope curriculum). Parent-teacher meetings were held at all LEAPS preschools once per academic year. NCHD officers conducted monthly preschool supervision visits supported by a supervision checklist to assess preschool quality and fidelity. Throughout the intervention, community youth leaders received refresher trainings and participated in youth development activities. Community youth leaders received a monthly stipend of 8000 Pakistani rupees (approximately US\$50), aligned with NCHD's salary structure for teachers. LEAPS preschools were closed from March to September, 2020, during the COVID-19 shutdown, at which time NCHD implemented LEAPS' emergency response plan to provide remote learning activities for the skill development of community youth leaders and workbooks for all LEAPS students.

Data collection was conducted by female Pakistani assessors residing locally, fluent in Sindhi, with at least a bachelor's level education. Most assessors had previous experience working in early child development trials. All assessors received a 10-day training at baseline plus a 1-day refresher before each data collection round. During each of the four survey rounds, a team of trained assessors compiled a list of all eligible children in each

village cluster using village birth registries verified through household visits. Assessors enrolled up to 11 child-caregiver dyads per village cluster, and if there were more than 11 eligible participant dyads, assessors used a random number generator to select study participants. Samples were independently selected at each survey round, and children were eligible to be re-enrolled at different survey rounds if they met all inclusion criteria. Children were assessed at their household, with breaks and refreshments as needed. Child assessments were conducted manually and then data were transferred electronically to tablets. Children and caregivers were given a small appreciation gift (eg, book or toy) after the session.

Outcomes

Children's school readiness was measured by the International Development and Early Learning Assessment (IDELA), which was previously validated in Pakistan.²³ It comprises the total score, which is the primary outcome of interest for this study, and subdomains of emergent numeracy, emergent literacy, socio-emotional development, and motor skills, which are secondary outcomes in this study. Other secondary outcomes were children's executive functions, assessed using a battery of tasks similarly adapted for use in Pakistan.²¹ Tasks included Corsi block-tapping, an assessment of children's working memory; knock and tap, a measure of the child's inhibitory control; peg tapping, which also assesses inhibitory control; Dimensional Change Card Sort, a measure of children's cognitive flexibility; and Head-Toes-Knees-Shoulder, a measure of general executive functioning skills (Cronbach's α in appendix 2 p 42).

Statistical analysis

Power was calculated based on changes in average child outcomes using representative population surveys.²⁴ An intraclass correlation coefficient of 0.175 was calculated based on the results of the LEAPS efficacy trial²¹ using the control group's total IDELA score at baseline. With three steps and a fixed number of village clusters ($n=99$; $n=33$ randomised per step), we calculated that at least eight children per village cluster per round, or 792 children per survey round, were needed to detect a 0.30 SD effect size with an alpha of 0.05 and 80% power. To achieve adequate power and account for refusal rates, we aimed to enrol 11 child-caregiver dyads per village cluster, totalling 1089 child-caregiver dyads per round, allowing for up to 30% refusal or under-recruitment rate in each round of the data collection.

Baseline balance was tested between randomised groups to see whether the randomisation was successful. Sociodemographic characteristics at the child, caregiver, and household levels were summarised across study groups. The primary outcome, continuous IDELA score, was calculated for each individual. Graded response

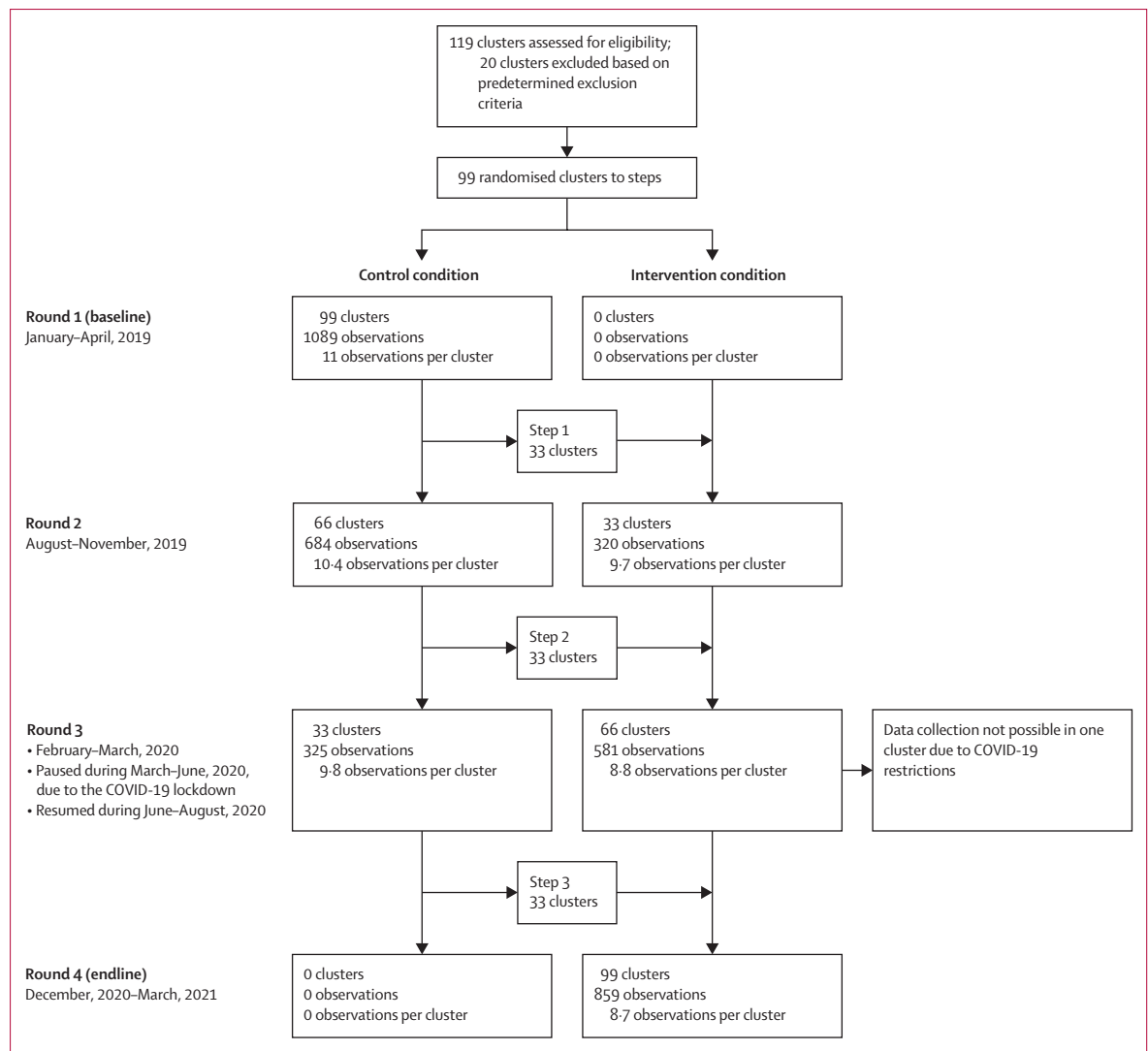


Figure 2: Study flow diagram

Observations per cluster represents the mean number of observations of children evaluated for the outcomes per cluster.

models were used to compute parameter estimates for IDELA scores.

Internal consistency for all outcomes was examined by computing Cronbach's α . Pearson correlation coefficients were calculated, and Bland–Altman plots were constructed to examine inter-rater agreement of measurements by assessors compared with measurements by the evaluation team field lead (SH).

All IDELA scores were computed through a Graded Response Model,²⁵ a specific application of Item Response Theory designed to handle ordinal responses and model the relationship between a child's latent ability in school readiness and their performance on IDELA test items. Internal consistency was adequate, with a Cronbach's α of 0.89 for the IDELA total score and ranging from 0.59 (motor skills) to 0.75 (emergent literacy) across subdomains (appendix 2 p 42).

Bland–Altman plots and Pearson correlation coefficients indicated good inter-rater agreement between assessors and the evaluation team field lead (SH) for all assessments, with Bland–Altman plots depicting low bias and narrow limits of agreement, and Pearson correlations ranging from $r=0.99$ to $r=1.00$ (appendix 2 p 43).

An intention-to-treat analysis was done following the analytical framework presented by Hussey and Hughes²⁶ for analysis of stepped-wedge cluster-randomised trials using a linear mixed model to produce the standard mean difference (SMD) in IDELA scores for LEAPS intervention versus control conditions. The linear mixed models included a fixed effect for stratified randomisation group, a random effect for cluster, and a fixed effect for survey round (appendix 2 p 40). We conducted sensitivity analyses outlined by Hussey and Hughes²⁶ to examine the robustness of the basic stepped-wedge cluster-randomised

	Clusters randomised to receive LEAPS intervention in step 1	Clusters randomised to receive LEAPS intervention in step 2	Clusters randomised to receive LEAPS intervention in step 3	Overall
Number of clusters	33	33	33	99
Number of clusters per district				
Dadu	2	2	2	6
Khairpur	9	9	10	28
Naushahro Feroze	19	19	18	56
Sukkur	3	3	3	9
Child characteristics				
Number of children	361	366	362	1089
Sex				
Girls	182 (50.4%)	188 (51.4%)	156 (43.1%)	526 (48.3%)
Boys	179 (49.6%)	178 (48.6%)	206 (56.9%)	563 (51.7%)
Age, years	5.0 (0.4)	5.0 (0.4)	5.1 (0.4)	5.0 (0.4)
Birth order	3.6 (2.3)	3.9 (2.6)	3.7 (2.4)	3.7 (2.4)
Private or non-governmental organisation preschool enrolment	3 (0.8%)	2 (0.6%)	6 (1.7%)	11 (1.0%)
NCHD or government feeder primary school enrolment	218 (60.4%)	227 (62.0%)	257 (71.0%)	702 (64.5%)
Main caregiver characteristics				
Mother is the main caregiver	355 (98.3%)	360 (98.4%)	356 (98.3%)	1071 (98.3%)
Age, years	33.4 (6.6)	34.0 (6.5)	33.6 (6.5)	33.7 (6.5)
Education level				
No formal schooling	302 (83.7%)	309 (84.4%)	296 (81.8%)	907 (83.3%)
Some primary school, up to grade 5	19 (5.3%)	17 (4.6%)	18 (5.0%)	54 (5.0%)
Completed primary school, grade 5	40 (11.1%)	40 (10.9%)	48 (13.3%)	128 (11.8%)
Occupation				
Housewife	286 (79.2%)	280 (76.5%)	288 (79.6%)	854 (78.4%)
Handicraft	56 (15.5%)	57 (15.6%)	52 (14.4%)	165 (15.2%)
Daily paid worker	12 (3.3%)	21 (5.7%)	18 (5.0%)	51 (4.7%)
Main provider characteristics				
Father is the main provider	354 (98.1%)	361 (98.6%)	354 (97.8%)	1069 (98.2%)
Education level				
No formal schooling	160 (44.3%)	169 (46.2%)	162 (44.8%)	491 (45.1%)
Some primary school, up to grade 5	12 (3.3%)	21 (5.7%)	14 (3.9%)	47 (4.3%)
Completed middle school and some middle or lower secondary school, grade 5 up to grade 10	85 (23.6%)	81 (22.1%)	77 (21.3%)	243 (22.3%)
Completed lower secondary school or higher, grade 10	97 (26.9%)	88 (24.0%)	100 (27.6%)	285 (26.7%)
Main occupation				
Farmer	112 (31.0%)	106 (29.0%)	108 (29.8%)	326 (29.9%)
Daily paid worker	71 (19.7%)	79 (21.6%)	87 (24.0%)	237 (21.8%)
Industrial worker	81 (22.4%)	69 (18.9%)	47 (13.0%)	197 (18.1%)
Household characteristics				
Family size	10.7 (5.2)	10.6 (6.3)	12.3 (9.2)	11.2 (7.2)
Number of children in household	5.0 (2.3)	5.2 (2.5)	5.0 (2.3)	5.1 (2.4)
Family structure				
Joint or extended family	244 (67.6%)	220 (60.1%)	240 (66.3%)	704 (64.7%)
Nuclear family	117 (32.4%)	146 (39.9%)	122 (33.7%)	385 (35.4%)
Household wealth quintiles	2.9 (1.3)	3.1 (1.5)	3.0 (1.5)	3.0 (1.4)

(Table 1 continues on next page)

	Clusters randomised to receive LEAPS intervention in step 1	Clusters randomised to receive LEAPS intervention in step 2	Clusters randomised to receive LEAPS intervention in step 3	Overall
(Continued from previous page)				
Household assets				
Private, reliable source of drinking water	348 (96.4%)	352 (96.2%)	344 (95.0%)	1044 (95.9%)
Toilet; latrine with flush system	97 (26.9%)	126 (34.4%)	142 (39.2%)	365 (33.5%)
Use electricity for lighting	261 (72.3%)	298 (81.4%)	297 (82.0%)	856 (78.6%)
Own at least one television	112 (31.0%)	151 (41.3%)	153 (42.3%)	416 (38.2%)
Household food security				
Food secure	175 (48.5%)	183 (50.0%)	172 (47.5%)	530 (48.7%)
Mildly to moderately food insecure	125 (34.6%)	139 (38.0%)	145 (40.1%)	409 (37.6%)
Severely food insecure	61 (16.9%)	44 (12.0%)	45 (12.4%)	150 (13.8%)

Data are n, n (%), or mean (SD). Preschool-aged children can be accommodated in the multigrade classroom of the NCHD feeder school. The NCHD does not implement an early childhood care and education programme. It does not have an early childhood care and education curriculum nor receive resources (eg, developmentally appropriate learning materials) for these children. Some multigrade NCHD feeder primary schools might be for preschooler and first and second grade children, whereas others might be for all primary school aged children. The mean household wealth quintiles are indicated by 1 representing the lowest wealth quintile and 5 representing the highest wealth quintile in the study population. LEAPS=Youth Leaders for Early Childhood Assuring Children are Prepared for School. NCHD=National Commission for Human Development.

Table 1: Baseline characteristics of children aged 4.5–5.5 years in LEAPS baseline survey overall and stratified by randomisation group

	LEAPS intervention effect in standardised mean difference (95% CI)	p value	Intraclass correlation coefficient
Primary outcome			
Total IDELA score	0.30 (0.20 to 0.40), n=3852	<0.0001	0.13
Secondary outcomes			
IDELA domain scores			
IDELA emergent numeracy score	0.32 (0.22 to 0.41), n=3852	<0.0001	0.08
IDELA emergent literacy score	0.22 (0.13 to 0.32), n=3852	<0.0001	0.14
IDELA socioemotional score	0.21 (0.12 to 0.30), n=3852	<0.0001	0.09
IDELA motor score	0.06 (−0.03 to 0.14), n=3852	0.19	0.07
Child executive function skills			
Corsi block-tapping score	0.08 (−0.02 to 0.19), n=3856	0.12	0.03
Knock-tap score	0.07 (−0.03 to 0.18), n=3856	0.17	0.05
Peg tapping score	−0.003 (−0.110 to 0.100), n=3856	0.96	0.04
Dimensional Change Card Sort score	0.22 (0.12 to 0.33), n=3856	<0.0001	0.04
Head Toes Knees Shoulders score	0.02 (−0.09 to 0.12), n=3856	0.78	0.06

LEAPS=Youth Leaders for Early Childhood Assuring Children are Prepared for School. IDELA=International Development and Early Learning Assessment.

Table 2: Effect of LEAPS intervention on primary and secondary outcomes across the whole trial

trial model assumptions (appendix 2 p 40). Not specified in the trial protocol, to show the difference in effects of LEAPS for round three (after the onset of COVID-19 and school closures) versus round two (before the onset of COVID 19), a fixed-effect interaction term between treatment and time to the linear mixed model was added. Analyses were done using Stata version 16. We also did a cost analysis to estimate cost projections for a province-wide expansion of LEAPS (appendix 2 p 44).

Role of the funding source

The funders had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

LEAPS was implemented in 91 of 99 village clusters from Dec 3, 2018, to June 30, 2021. In eight clusters, a LEAPS preschool could not be set up (eg, due to a lack of space or unable to recruit a community youth leader). For the intervention programme, the average enrolment of children in a LEAPS preschool was 19 (SD 3, range 12–20). A total of 3858 children (n=3852 with complete IDELA data) were assessed across the four survey rounds conducted between Jan 1, 2019, and March 31, 2021, to test the effect of the programme (figure 2).

Child, main caregiver, main provider, and household characteristics at the baseline survey stratified by randomised group is provided in table 1. All characteristics appeared to be balanced by randomisation group at baseline. A majority of children at baseline (702 [64.5%] of 1089) were enrolled in NCHD feeder primary schools or government primary schools (outside of their own village), in which children typically attend multiage primary school classes with older siblings with little or no ECCE provision. Child characteristics in each of the follow-up survey rounds are presented in appendix 2 (pp 17–20). Among children surveyed in LEAPS intervention village clusters, LEAPS programme coverage was 68.4% overall (1204 LEAPS attendees of 1760 children assessed), with 76.6% in round two (normal operation; 245 of 320), 59.0% in round three (remote learning and full school closure due to COVID-19; 343 of 581), and 71.4% in round four (endline partial school closure due to COVID-19; 613 of 859).

The estimated effects of the LEAPS intervention on children's school readiness and executive functioning skills are shown in table 2. LEAPS improved IDELA overall scores by 0.30 SMD (95% CI 0.20 to 0.40; $p<0.0001$) as compared with the control condition. Effect sizes (SMD) for IDELA subdomains were 0.32 for

	Round 2: effect of LEAPS before onset of COVID-19 pandemic (95% CI)	Round 3: effect of LEAPS during and after implementation of COVID-19 response strategies (95% CI)	Difference in effect of LEAPS for round 3 versus round 2 (95% CI); Interaction term intervention × round	p value
Primary outcome				
Total IDELA score	0.37 (0.22 to 0.52)	0.05 (−0.10 to 0.20)	−0.33 (−0.51 to −0.15)	0.0004
Secondary outcomes				
IDELA domain scores				
IDELA emergent numeracy score	0.37 (0.24 to 0.51)	0.12 (−0.01 to 0.26)	−0.25 (−0.42 to −0.08)	0.0043
IDELA emergent literacy score	0.28 (0.14 to 0.42)	0.04 (−0.10 to 0.18)	−0.24 (−0.42 to −0.07)	0.0060
IDELA socioemotional score	0.26 (0.14 to 0.38)	0.02 (−0.10 to 0.14)	−0.23 (−0.39 to −0.08)	0.0026
IDELA motor score	0.18 (0.05 to 0.30)	−0.06 (−0.19 to 0.06)	−0.24 (−0.40 to −0.09)	0.0021
Child executive function skills				
Corsi block-tapping score	0.12 (−0.02 to 0.26)	0.03 (−0.12 to 0.17)	−0.09 (−0.29 to 0.10)	0.34
Knock-tap score	0.20 (0.05 to 0.35)	−0.08 (−0.24 to 0.07)	−0.29 (−0.48 to −0.09)	0.0043
Peg tapping score	0.07 (−0.09 to 0.23)	−0.12 (−0.28 to 0.03)	−0.19 (−0.40 to 0.01)	0.061
Dimensional Change Card Sort score	0.37 (0.22 to 0.52)	0.09 (−0.07 to 0.24)	−0.28 (−0.48 to −0.08)	0.0052
Head Toes Knees Shoulders score	0.27 (0.12 to 0.43)	−0.19 (−0.35 to −0.02)	−0.46 (−0.67 to −0.25)	<0.0001

Effect in standardised mean difference (95% CI) of LEAPS intervention on primary and secondary outcomes in round 2 before onset of COVID-19 pandemic and round 3 during and after implementation of COVID-19 response strategies. We kept observations from round 2 and round 3 only (n=1907) and added a fixed-effect interaction term between treatment and time to the linear mixed model to estimate the effect of LEAPS in round 2 and round 3. LEAPS=Youth Leaders for Early Childhood Assuring Children are Prepared for School. IDELA=International Development and Early Learning Assessment.

Table 3: Effect of LEAPS intervention on primary and secondary outcomes in rounds 2 and 3

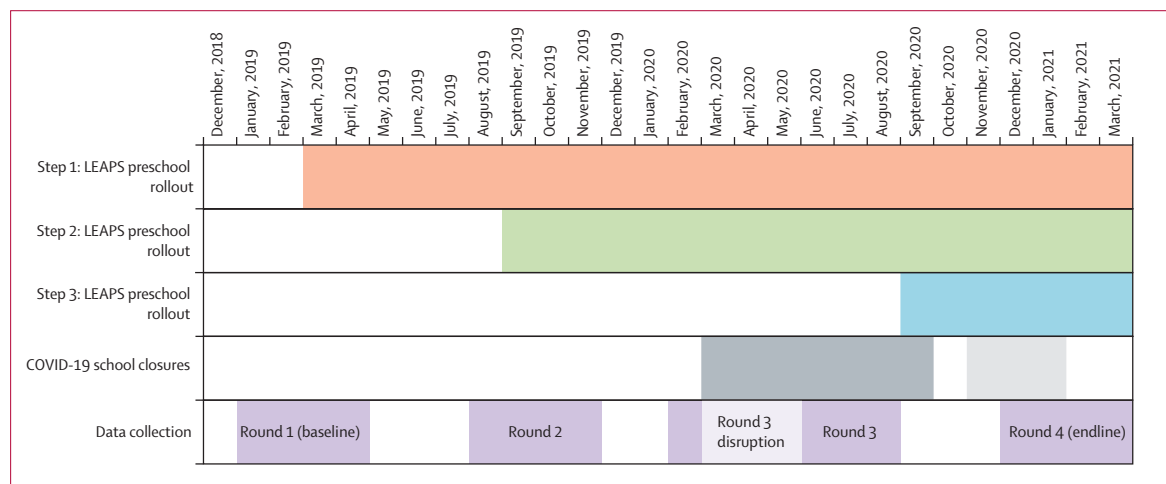


Figure 3: Timeline of LEAPS preschool rollout, COVID-19 school closures, and data collection activities

The dark grey cells represent full school closure between March 3, 2020, and Sept 27, 2020, due to the COVID-19 pandemic. The light grey cells represent partial school closure between Nov 26, 2020, and Jan 11, 2021, due to the COVID-19 pandemic. Round 3 data collection was disrupted due to the COVID-19 pandemic restrictions. LEAPS=Youth Leaders for Early Childhood Assuring Children are Prepared for School.

numeracy skills (0.22 to 0.41; $p<0.0001$), 0.22 for literacy skills (0.13 to 0.32; $p<0.0001$), 0.21 for socio-emotional development (0.12 to 0.30; $p<0.0001$), and 0.06 for motor skills (−0.03 to 0.14; $p=0.19$). With respect to executive functioning skills, only cognitive flexibility showed improvements as shown by the dimensional change card sort score (SMD 0.22 [95% CI 0.12 to 0.33]; $p<0.0001$; table 2; appendix 2 p 21).

We conducted multiple sensitivity analyses for the linear mixed models. Sensitivity analyses including

per-protocol analysis, a fixed-effect model for clusters, a random-effect interaction between treatment and clusters, a random-effect interaction between time and clusters, a generalised estimating equation model, and a multivariate model all produced the same results as the linear mixed model did (appendix 2 pp 22–39). However, we found that the effectiveness of LEAPS differed with respect to survey round (table 3). The effect of LEAPS on total IDELA scores was higher in round two (SMD 0.37 [95% CI 0.22 to 0.52]; $p<0.0001$)

before the onset of the COVID-19 pandemic as compared with round three (SMD 0·05 [−0·10 to 0·20]; $p=0·53$), which was conducted during and after the implementation of COVID-19 response strategies (figure 3). There was also a decline in the effectiveness of LEAPS from round two to round three on all secondary outcomes, except Corsi block-tapping and peg tapping score. Scores for motor skills were improved in round two (SMD 0·18 [0·05 to 0·30], $p=0·0046$), though effects were attenuated over the full trial. A descriptive summary of changes in primary and secondary outcomes by survey round and intervention status is also presented in appendix 2 (p 41).

The estimated total cost-per-child for 1 year of LEAPS attendance when implemented at province-wide scale was US\$106 (16 539 Pakistani rupees) when accounting for start-up costs (ie, initial trainings), and \$95 (14 821 rupees) when start-up costs were excluded (appendix 2 pp 44–45).

Discussion

As the NCHD is charged with the provision of basic education and training services in areas of Pakistan where the Ministry of Federal Education and Professional Training is unable to operate, LEAPS was reaching highly vulnerable children. Despite interruptions in ECCE due to COVID-19 school closures, LEAPS benefitted the primary outcome of children's composite school readiness scores measured as the total IDELA score. No overall effects were observed on children's motor skills or executive function skills, except for cognitive flexibility. However, benefits to all outcomes were stronger in round two of data collection (before the onset of COVID-19), relative to round three (during the COVID-19 pandemic), except for the inhibitory control assessments.

ECCE programmes have shown short-term benefits to children's early development and school readiness, as well as longer-term benefits in the life course with respect to their physical and mental health, educational achievement, and earnings.¹³ However, access to ECCE remains poor with inequities in populations. LEAPS, a non-formal community-based ECCE programme led by trained female youth, was designed in response to closing these gaps by operating in communities with no formal ECCE programme and concurrently promoting youth development and economic empowerment for female youth who have fewer opportunities than their male counterparts.²² Youth-led models, employing peer mentoring or cross-age mentoring (working with younger mentees), have been implemented to promote low risk-behaviours and support health and nutrition education activities.^{19,27} LEAPS showed that youth could effectively deliver ECCE. While the analysis of the youth outcomes for this larger-scale implementation of LEAPS are ongoing, the pilot study showed gains in professional skills development, sense of self-worth, and an interest in developing a career in teaching.¹⁶

Reviews have reported both formal and non-formal or community-based ECCE programmes yield medium-to-high impacts on children's cognitive development and small impacts on children's psychosocial development.^{14,28} The LEAPS programme effect size was relatively small, but it is noteworthy that 9 months of implementation was disrupted due to complete or partial lockdowns of preschool as a result of the COVID-19 pandemic, reducing the number of in-person learning days for young children as well as teaching experience of the newly trained community youth leaders. Notably, the effect on the primary outcome of total IDELA scores, capturing school readiness skills in early literacy and numeracy, motor skills, and socioemotional skills, that was observed during round two data collection before onset of the pandemic is much more substantial than the effect size observed during round three of data collection after the outbreak of COVID-19. Studies of participation in ECCE for young children in LMICs²⁹ and in high-income settings³⁰ found benefits to school readiness skills.

Irrespective of the nature of the programme, predictors of programme effectiveness include structural quality (eg, classroom size, teacher-to-child ratio, and various learning materials) and process quality (eg, responsive teacher-child interactions, and classroom emotional climate).^{14,28} Global systematic reviews of ECCE generally show positive associations with quality improvements and child outcomes.¹ Although this study did not investigate the quality of LEAPS ECCE, inputs to support quality were considered. One factor contributing to poor access and low quality of ECCE is an early childhood workforce gap with staff often poorly paid with inadequate training and support. The LEAPS programme integrated a youth-mentoring programme to provide supervision and feedback to youth educators after receiving their ECCE training. Investments in preschool teacher training and support have been identified as crucial to supporting quality in ECCE programmes; for example, in Ghana, an evaluation of a 1-year teacher training and coaching intervention yielded benefits to children's early academic skills and socio-emotional development.³¹ A second factor influencing quality is low-demand for ECCE, which can emerge from a lack of parental engagement in early learning programmes.³² LEAPS fostered community engagement through community sensitisation and awareness sessions about ECCE, community participation in recruiting the community youth leaders, community identified and donated space for the preschool, community volunteers to support the running of preschools (eg, community participation in celebration days, and creating learning resources), and parent-teacher meetings. Future analyses will need to investigate how, if at all, quality in LEAPS is associated with child outcomes.

With respect to cost, we estimate the total cost of delivering a holistic child-centred curriculum at scale in

Sindh per child per year at \$106 (16 539 rupees) with most costs contributing to preschool materials and operations. Cost per child per year drops to \$95 (14 821 rupees) when excluding start-up costs. There were limitations in this analysis, including assumptions about time to scale-up, and exclusion of participant costs, such as time, which could provide a more comprehensive estimate of actual social costs. In India, the integration of a 4-month early childhood numeracy curriculum in a non-formal preschool system was \$316 for a group of six children with approximately two-thirds of the costs for learning materials.³³ A review of costs of implementing ECCE services in LMICs, reports the average cost per child ranges from \$21 to \$2945, with a median cost of \$289 per child and year.¹³ In contrast to investing, the Pakistan Human Capital Report notes the cost of inaction on medium coverage of ECCE services in the country is estimated at \$4.7 billion for each preschool cohort that fails to receive medium coverage.²² Further analysis is underway to investigate the effect of LEAPS on youth development and to study the implementation features associated with programme effectiveness.

This Article reports the short-term benefits of ECCE delivered by youth living in a disadvantaged setting. However, future studies on this cohort offer two key opportunities. First, it is imperative that longitudinal studies should evaluate the medium-to-longer term benefits to child and youth health and development. Previous evaluations from studies in the USA¹⁰ and Türkiye¹¹ have shown a host of health outcomes for adults who were exposed to ECCE services in childhood. LEAPS offers an opportunity to assess health and development of two generations over time. Second, micronutrient deficiencies, anaemia, and undernutrition persist in the preschool age group. Whereas health services are primarily responsible for nutrition interventions in the first 1000 days, ECCE services such as LEAPS might offer a platform to implement nutrition interventions at scale during the next 1000 days, thereby supporting holistic nurturing care inputs.¹ The LEAPS intervention could also have spill-over effects on children who did not attend LEAPS through communication between mothers or caregivers as well as play between children in the community which further justified conducting a population-level survey.

The stepped-wedge cluster-randomised design, the culturally adapted and standardised school readiness and executive function measures, and the reliable administration of assessments are all strengths of the study. However, there are several limitations. First, with respect to programme design, a few communities were unable to provide community space for a LEAPS preschool or were unable to identify a female community youth leader who met the education and age criteria, thus reducing the number of final clusters that implemented the intervention. Second, generalisability of these findings is limited to contexts with

large numbers of young children without access to ECCE and at risk of inadequate nurturing care inputs, as well as populations with a high youth prevalence who have low education, training, and employment opportunities. Given that not all students included in the surveys attended the LEAPS programme, the magnitude of the programme effects might have been attenuated.

Nonetheless, in many disadvantaged settings, participation in ECCE remains low and accelerating progress to meet the SDG target on early childhood care, development, and education requires both innovations that address the existing realities of communities and inequities on the ground, and a host of partners to support implementation. In well-resourced settings, even if demand for ECCE is high, recruitment and retention of a well trained workforce is challenging. In California, USA, a promising pilot of the Early Care and Education Pathways to Success model showed the value of a youth apprenticeship model for youth themselves as well as addressing the workforce challenge.³⁴ Such models have the potential to yield a triple dividend for young people by supporting youth now, future opportunities, and next generation (as parents).¹⁸ The LEAPS findings indicate a community-based youth-led ECCE approach is a promising strategy to expand access to high-quality ECCE and promote children's school readiness in disadvantaged communities while providing employment for female youth, which could be feasibly delivered on a larger scale. It is recognised that to achieve both scale and impact, systems that support early child development must address challenges, such as workforce development, financing, and community demand.¹⁵ Going forward, evidence generated on early childhood development and learning interventions must now address these challenges and measure progress to ensure the youngest children's health and development needs are holistically met through multisectoral actions.

Contributors

AKY conceptualised the study. AKY, CRS, SS, EEF, CRR, GF, and LAP developed the study design and implementation strategies. EEF, SS, KR, and QD developed the study protocols and data collection materials with inputs from AKY and SB. SS coordinated the implementation of the study and oversaw trainings with support from EEF, KR, and SB. SH oversaw data collection and quality assurance with support from SS, EEF, and KR. CRS and QD developed the plan of analysis and did the statistical analysis with inputs from AKY, EEF, SB, GF, CRR, and LAP. CS, QD, EF, and AY accessed and verified the data. All authors contributed to the interpretation of findings. AKY drafted the manuscript with support from EEF, KR, and QD. All authors critically reviewed drafts of the manuscript. AKY is the guarantor for the research. All authors had access to the raw data and all authors agreed to submit the manuscript for publication. The Position Statements of all authors in this global collaboration can be found in appendix 2 (pp 46–48).

Declaration of interests

We declare no competing interests.

Data sharing

Data used in this analysis can be accessed at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/VUETQC>. The

published trial protocol can be accessed at <https://trialsjournal.biomedcentral.com/articles/10.1186/s13063-021-05518-9>.

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